Can Photos Prove an Aero-Engine Blade Is Damaged?

Finding 3D damage from a CAD model and a few photos, and knowing when the photos are not enough

Afeefa Azam | Khalifa University

1. The problem

A dark spot in a photo of a blade can have **four different causes**. Only the first is physical damage, yet in a photo they can look almost identical: a 0.1 mm dent changes the image only slightly, while moving the camera by 1° changes thousands of pixels. A computer that simply looks for "something different" raises many false alarms.

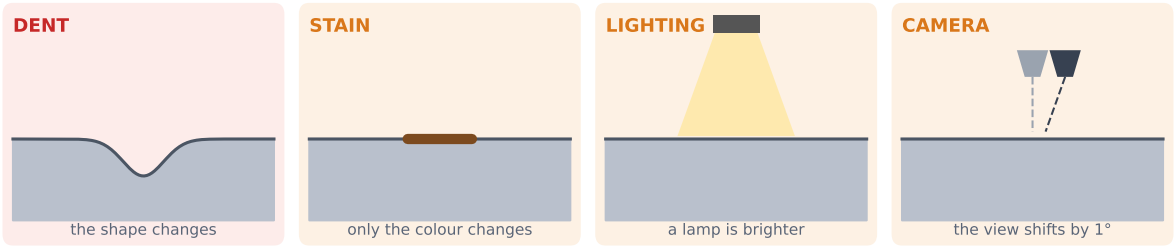


Figure 1. One dark spot, four possible causes.

2. The goal

To build a method that reports blade damage in 3D **only when the photos can prove it**, and otherwise says **"Not enough evidence."**

Objectives: (1) Separate real damage from stains, lighting changes and camera errors. (2) Locate the damage on the 3D blade and estimate its depth. (3) Recognise when the photos are insufficient, and abstain.

3. Methodology: the three-step evidence test

The computer draws (renders) the healthy blade from its CAD model with a differentiable renderer [6] and compares the drawing with the photos. Step 1 gives a mismatch score with no damage, $L_{\text{no damage}}$; step 2 lets the 3D mesh dent at suspect spots and gives L_{damage}. The threshold is set on healthy and stained blades so that only about 5% of undamaged blades are flagged.

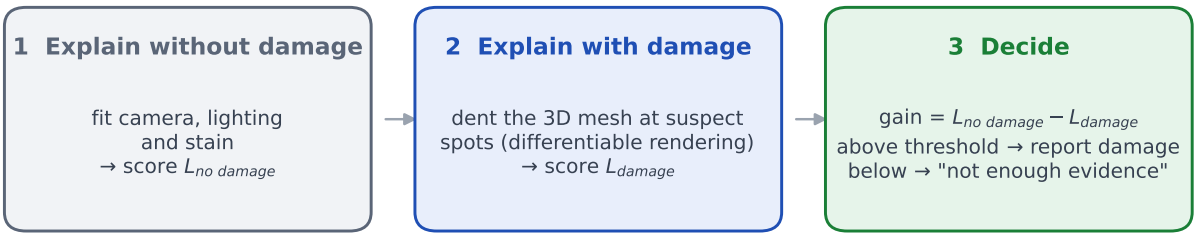


Figure 2. The evidence test.

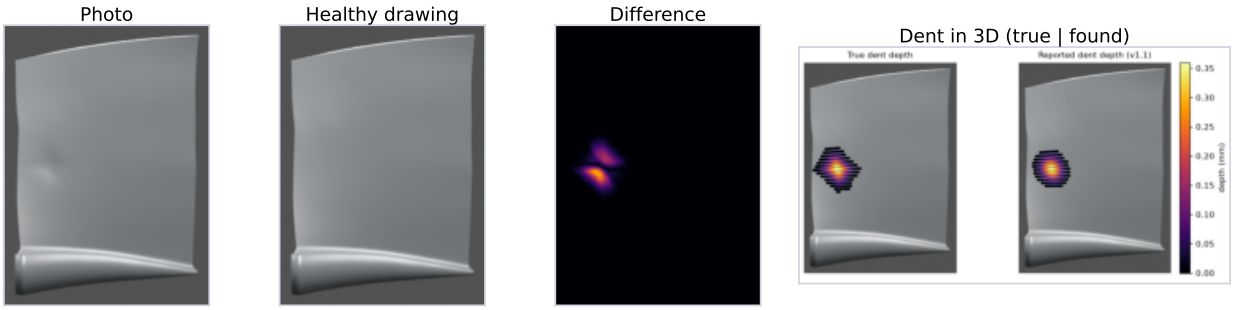


Figure 3. A real output: photo, healthy drawing, their difference, and the dent found in 3D (0.35 mm; true 0.36 mm).

4. The specialists and judge: which damage is it?

To identify the damage type, four physical models (*specialists*) each try to explain the photos at the damaged spot. A *judge* compares how well each one fits: a clear winner gives the type; if two fit about equally well, it says "type uncertain" instead of guessing. The judge only labels damage already found by the evidence test, so it cannot add false alarms. (Idea adapted from multi-agent checking in MedGuards.)

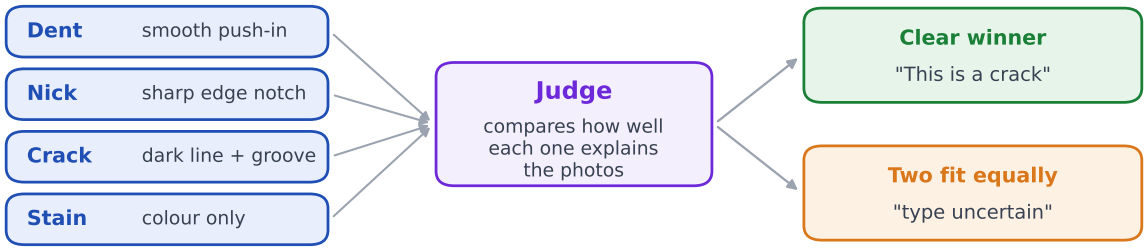
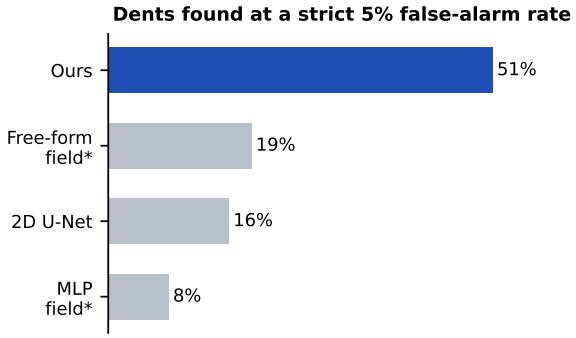


Figure 4. Specialists and judge.

5. Preliminary results (about 1,350 computer-rendered cases)



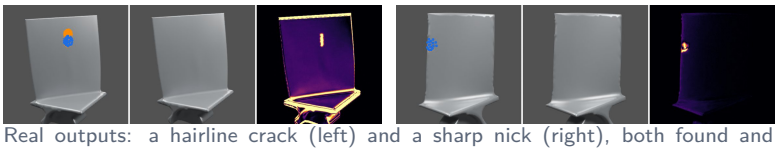
*our versions of the CAD-based method [4]

100% crack type correct (22/22)

9/15 nick type correct, +6 "uncertain"

0.7 mm median 3D location error

51%→4% false alarms on unseen blade designs (v1.1 fix)



Real outputs: a hairline crack (left) and a sharp nick (right), both found and correctly typed. Orange = true damage, blue = found.

6. The identifiability finding: why depth is the hard part



On an unseen industrial blade, a **0.64 mm** dent was found in the right place but measured as **0.06 mm**: in two ordinary photos, a small sharp dent of very different depths can produce nearly identical pixels. Our experiments show depth is **not reliably identifiable** from two passive photos: on new designs, found dents are measured at a median 60% of their true depth, and more cameras or depth ranges did not fix it. So the system reports *where* the damage is, which is reliable, and treats depth as the open problem. An author of PIAD [3], in personal correspondence, independently suggested that studying identifiability and abstaining is a meaningful contribution in itself.

7. Next step: real blades

1. **3D print** 10 stainless-steel blades from our CAD models.
2. **Add damage:** ball bearings (dents), files (nicks), paint (stains).
3. **Photograph** with 2–7 cameras, ArUco markers for camera positions, and grazing (low-angle) light.
4. **Scan or gauge** each defect to measure its true depth in mm.
5. **Compare:** run the frozen pipeline and compare its results with the scan.

References

[1] M. A. Mohammed Eltoum et al., BladeSynth, *Scientific Data*, 2025. [2] Zhou et al., PAD, NeurIPS 2023. [3] Yang et al., PIAD, CVPR 2025. [4] Overdulve et al., CADSplat, arXiv 2026. [5] Costanzino et al., SiM3D, ICCV 2025. [6] Laine et al., Modular primitives for high-performance differentiable rendering (nvdiffrast), 2020.